Tanmay Kadlag

Machine Learning Engineer

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Professional Summary

Results-driven Machine Learning Engineer with 3+ years at Deloitte, designing and deploying end-to-end ML pipelines and predictive models for large-scale FMCG clients. Proficient in Python, PySpark, Scala, and Databricks, with hands-on experience across the full ML lifecycle — data preprocessing, model training, tuning, deployment, and monitoring. Proven record of improving model accuracy by up to 30% and reducing prediction errors by 15% on 500M+ record datasets.

Technical Skills

ML & Data Science: Predictive Modeling, Sales Forecasting, Clustering, Model Tuning, Model Deployment, Retraining Pipelines, CI/CD for ML
Languages: Python, Scala, SQL, Spark SQL, PySpark, Apache Spark
Cloud & Platforms: Databricks, Azure Data Factory (ADF), ADLS Gen2, Delta Lake, Databricks Workflows, Google BigQuery
Data Engineering: ETL/ELT, Medallion Architecture, Delta Lake, Star/Snowflake Schema, Change Data Capture (CDC)
Visualization & Monitoring: Power BI, Looker Studio, Report Automation, Model Performance Dashboards
Tools: Git, Office 365

Work Experience

Deloitte — Technology & Transformation, Innovation Team May 2024 – Present
Machine Learning Engineer (Big Data) *Pune, India*

- Led end-to-end industrialization of a predictive analysis pipeline using Azure Data Factory for a national-scale deployment, achieving full platform integration and improving **model deployment efficiency by 30%**.
- Engineered a comprehensive ETL framework processing **500M+ records/month** in Scala and Python, feeding ML models deployed via Databricks Workflows for enhanced sales prediction accuracy.
- Designed and launched a new ML model from scratch with the innovation team, boosting **model performance by 10%** through advanced clustering techniques and iterative model training.
- Streamlined model integration into operational workflows via tuning and retraining pipelines, increasing **pipeline efficiency by 20%**.
- Optimized FMCG sales forecasting using ML algorithms, reducing **prediction errors by 15%**; built and managed **2TB+ data pipelines** using Delta Lake and PySpark.

Deloitte — Cloud Engineering Jul 2022 – May 2024
Analyst — ML & Data Engineering *Mumbai, India*

- Enhanced predictive analysis accuracy by **30%** by preprocessing and feature-engineering large datasets using PySpark and Scala for churn prediction and sales recommendation models.
- Automated report generation using Scala, reducing **manual reporting effort by 30%** and delivering dynamic Excel-based performance dashboards for sales teams.
- Improved Power BI dashboard refresh speed by **25%** through backend query optimization, enabling real-time model performance and health monitoring.
- Streamlined ETL preprocessing workflows using Apache Spark, improving **data readiness by 40%** for accurate downstream sales forecasting models.
- Built traffic-light monitoring systems with performance thresholds to track model health and trigger automated retraining, ensuring consistent model reliability.

Projects

End-to-End FMCG Data Lakehouse | *Databricks, PySpark, Delta Lake, Medallion Architecture, CDC*

- Designed a scalable Data Lakehouse on Databricks to unify disparate FMCG data sources after a simulated merger, establishing a single source of truth from raw Amazon S3 ingestion.

- Implemented a **Medallion Architecture** (Bronze/Silver/Gold) with structured data quality progression and ACID-compliant Delta Lake transactions.
- Built advanced PySpark pipelines with CDC logic using `MERGE INTO` for efficient upserts and deduplication; designed a **Star Schema** in the Gold layer for optimised BI reporting.
- Integrated **Databricks Genie** for natural language, ad-hoc business queries — enabling self-service analytics for non-technical stakeholders.

GCP Student Placement Analytics Pipeline | *GCP, BigQuery, SQL, Looker Studio*

- Architected and deployed a cloud-native ELT pipeline to analyze employability factors for **5,000+ engineering students**, surfacing data-driven placement insights.
- Discovered the **Backlog Cliff Effect** — even one active backlog reduced placement probability from 94% to 73%; found internship experience had a **2× stronger correlation** with high salary (>15 LPA) versus CGPA alone.
- Built a live, interactive Looker Studio dashboard with cross-filtering and dynamic insights connected directly to BigQuery materialized views.

Education

Visvesvaraya National Institute of Technology (VNIT) — Nagpur, India
Bachelor of Technology — Electrical and Electronics Engineering — *2018 – 2022*

Certifications

Machine Learning Specialization — Stanford University & DeepLearning.AI — Mar 2025
AI-900: Azure AI Fundamentals — Microsoft — Sep 2025
Databricks Fundamentals Accreditation — Databricks — Jul 2025
Artificial Intelligence Foundations: Thinking Machines — LinkedIn Learning — Aug 2024
Python for Data Science, AI & Development — IBM — Jun 2024
Data Science Methodology — IBM — May 2024

Achievements

JENESYS 2.0 Youth Exchange Program — Represented India — 2014
Inter-NIT Yoga Tournament Participant — VNIT Nagpur — 2019